Reg. No. | | | | | | | | | | | [14AE35]

THIRD SEMESTER B.TECH. EXAMINATION, DECEMBER - 2015

THERMODYNAMICS

Time: 3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions. Taking at least TWO questions from each part.
2. Assume any missing data suitably.

PART – A

1. a. Distinguish between:

i. Closed and open systems

ii. Intensive and extensive properties

iii. Thermal and mechanical equilibrium **06**

b. State zeroth law of thermodynamics. How does this form the basis for temperature measurement. **04**

c. Sir Isaac Newton proposed a temperature scale in 1709. On this scale, temperature was a linear function of Celsius scale. The reading on this at Ice point(0°C) and normal human body temperature(37°C) were 0°N and 12°N respectively. Obtain the relation between the Newton scale and Celsius scale. **04**

2. a. Derive an expression for:

i. Shaft work ii. Spring work **05**

b. Air in a cylinder at an initial volume of $0.01m^3$ and initial pressure $6MP_a$ expands following a quasi-static process given by $PV^{1.4}$=constant. If the final volume of the gas is $0.025\ m^3$, determine the work done by the gas. **06**

c. Bring out two similarities and two dissimilarities between heat and work. **03**

3. a. State first law of thermodynamics as applied to:

i. Closed system ii. Open system **04**

b. Modify steady flow energy equation for:

i. Adiabatic diffuser

ii. Compressor with negligible potential and kinetic energy change. **04**

c. A turbine operates under steady flow condition receiving steam at the following state: Pressure $1.2MP_a$, Temperature188°C, enthalpy 2785KJ/Kg, velocity 34m/s and elevation 3m. The steam leaves the turbine in the following state: pressure $20KP_a$, enthalpy 2515KJ/Kg, velocity 100m/s and elevation 0m. Heat is lost to the surrounding at the rate of 0.29KJ/s. If the steam flow rate is 0.42Kg/s, determine the power output from the turbine. **06**

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4. a. Represent schematically heat engine, heat pump and refrigerator. Give their performance. **03**

b. State Kelvin-Plank and Clausius statements of second law of thermodynamics and show that they are equivalent. **06**

c. A reversible heat engine absorbs 200J of heat from reservoir at temperature of thermal boiling point of water and rejects heat to a reservoir at the temperature of triple point of water. Find the heat rejected, the work done by the engine and thermal efficiency. **05**

PART - B

5. a. Describe working of the Carnot engine and show that $\eta = 1 - \frac{T_2}{T_1}$. Represent Carnot cycle in p-v and T-s diagram. **05**

b. State and prove Clausius inequality. **04**

c. A inventor claims to have developed a power cycle capable of delivering 415KJ of net work output for an energy input by heat transfer 1000KJ. The system undergoing cycle receives heat from a source at 500K and rejects heat to a sink at 300K. Determine if this is a valid claim. **05**

6. a. Define entropy and prove that it is a property of the system. **04**

b. For ideal gas undergoing finite change of state from 1to 2 derive an expression for change in entropy. **05**

c. Draw P-T diagram of pure substance indicating triple and critical points. **05**

7. a. Relate first law efficiency, Carnot efficiency and second law efficiency. **05**

b. Define: **04**

i. availability

ii. Effectiveness

c. Calculate increase in availability of air when heated reversibly from 40°C to 250°C at constant pressure of 2atm. Assume the environment to be at 20°C. **05**

8. a. Briefly explain different types of classification of fluid machinery. **06**

b. Explain degree of reaction and efficiency of fluid machinery. **04**

c. Briefly explain with example Impulse and reaction turbines. **04**